ACE Model (Subset Data)

Step 0: Install and load required packages

```
# we recommend running this is a fresh R session or restarting your current session
install.packages(
    "cmdstanr",
    repos = c('https://stan-dev.r-universe.dev', getOption("repos"))
)
library(cmdstanr)
# Check toolchains
check_cmdstan_toolchain()
# If toolcahin is not setup properly, follow the printed instructions, or visit
https://mc-stan.org/docs/cmdstan-guide/installation.html#cpp-toolchain
# Install cmdstan
cmdstanr::install_cmdstan()
# We also install the bayesplot package visualizing MCMC results
install.packages("bayesplot")
```

```
library(cmdstanr)
```

```
This is cmdstanr version 0.9.0

- CmdStanR documentation and vignettes: mc-stan.org/cmdstanr

- CmdStan path: /home/marklai/.cmdstan/cmdstan-2.40.0

- CmdStan version: 2.40.0
```

```
register_knitr_engine()
options(mc.cores = min(parallel::detectCores() - 1, 4)) # Use four CPU cores for
MCMC sampling, if available
library(bayesplot)
```

```
This is bayesplot version 1.16.0

- Online documentation and vignettes at mc-stan.org/bayesplot

- bayesplot theme set to bayesplot::theme_default()

   * Does _not_ affect other ggplot2 plots
```

```
   * See ?bayesplot_theme_set for details on theme setting
```

```
library(HDInterval)
library(ggplot2)
```

```
# Define function for HDI
hdi2 <- function(x, cred_mass = 0.90) {
    hdi_interval <- HDInterval::hdi(x, credMass = cred_mass)
    setNames(hdi_interval, c("hdi_lo", "hdi_hi"))
}
```

Step 1: Data import

```
set.seed(1234)
n_mz <- c(325, 60, 7, 78, 150, 40, 8, 38, 88)
n_dz <- c(202, 97, 19, 96, 112, 64, 46, 56, 66)
# Convert to long format data frame
dat_mz <- data.frame(
    twin1 = rep(rep(1:3, 3), n_mz),
    twin2 = rep(rep(1:3, each = 3), n_mz)
)
dat_dz <- data.frame(
    twin1 = rep(rep(1:3, 3), n_dz),
    twin2 = rep(rep(1:3, each = 3), n_dz)
)
dat_mz[] <- lapply(dat_mz, ordered, labels = c("normal", "overweight", "obese"))
dat_dz[] <- lapply(dat_dz, ordered, labels = c("normal", "overweight", "obese"))
dat_full <- rbind(
    data.frame(zygosity = "MZ", dat_mz),
    data.frame(zygosity = "DZ", dat_dz)
)
sub_n <- 100
dat <- dat_full[sample.int(nrow(dat_full), 100), ]
# dat <- dat_full
```

```
str(dat)
```

```
'data.frame':   100 obs. of  3 variables:
 $ zygosity: chr  "DZ" "DZ" "DZ" "DZ" ...
 $ twin1   : Ord.factor w/ 3 levels "normal"<"overweight"<..: 2 2 1 2 3 1 3 1
```

```
1 1 ...
 $ twin2   : Ord.factor w/ 3 levels "normal"<"overweight"<..: 2 1 2 1 2 1 2 1
2 1 ...
```

```
summary(dat)
```

```
      zygosity          twin1           twin2
 Length   :100   normal    :53   normal    :48
 N.unique :  2   overweight:29   overweight:41
 N.blank  :  0   obese     :18   obese     :11
 Min.nchar:  2
 Max.nchar:  2
```

```
tab_mz <- with(dat[dat$zygosity == "MZ", ], table(twin1, twin2))
tab_dz <- with(dat[dat$zygosity == "DZ", ], table(twin1, twin2))
```

Step 2: Compile the Stan model

```
//
// This Stan program estimates the polychoric correlation between two
// ordinal variables, and decompose based on an ACE model. This model
// assumes symmetry in the contingency table.
//
functions {
  #include "ace_fun.stan"
}
data {
  int<lower=0> k;  // number of categories
  array[k * (k + 1) %/% 2] int<lower=0> ltri_ns1;  // mz
  array[k * (k + 1) %/% 2] int<lower=0> ltri_ns2;  // dz
}
transformed data {
  int kprime = k * (k + 1) %/% 2;
  int N1 = sum(ltri_ns1);  // total N for mz
  real N1real = N1;  // convert from integer to real number
  int N2 = sum(ltri_ns2);  // total N for dz
  real N2real = N2;  // convert from integer to real number
}
parameters {
  real<lower=0, upper=1> scaled_rho1;
  real<lower=0, upper=1> scaled_rho2;
```

```
  ordered[k - 1] thres;
}
transformed parameters {
  real rho1 = scaled_rho1 * 2 - 1;
  real rho2 = scaled_rho2 * 2 - 1;
  // Implied probabilities for each cell
  vector<lower=0>[kprime] pi1 =
    vec_symm(polycor_to_prob(thres, rho1));
  vector<lower=0>[kprime] pi2 =
    vec_symm(polycor_to_prob(thres, rho2));
}
// The model to be estimated.
model {
  // prior
  scaled_rho1 ~ beta(1.5, 1.5);
  scaled_rho2 ~ beta(1.5, 1.5);
  thres ~ student_t(4, 0, 2.5);
  // likelihood
  ltri_ns1 ~ multinomial(pi1);
  ltri_ns2 ~ multinomial(pi2);
}
generated quantities {
  // Compute a^2, c^2, and e^2 from rho
  real h2 = 2 * (rho1 - rho2);
  real c2 = 2 * rho2 - rho1;
  real e2 = 1 - rho1;
  // Intialize X^2 for each group
  real X2_1_samp = 0.0;
  real X2_2_samp = 0.0;
  // Initialize X^2 > X^2_rep indicator (for posterior predictive p)
  int<lower=0, upper=1> X2_1_gt;
  int<lower=0, upper=1> X2_2_gt;
  // Simulate new cell counts
  array[kprime] int ltri_ns1_tilde = multinomial_rng(pi1, N1);
  array[kprime] int ltri_ns2_tilde = multinomial_rng(pi2, N2);
  {
    vector[kprime] N1pi1 = N1real .* pi1;
    vector[kprime] N2pi2 = N2real .* pi2;
    // initialize X^2_rep for each group
    real X2_1_rep = 0.0;
    real X2_2_rep = 0.0;
    for (idx in 1:kprime) {
      real ns1_exp = N1pi1[idx];
```

```
      real ns2_exp = N2pi2[idx];
      // loop over each cell, and add the discrepancy to X2
      X2_1_samp += (ltri_ns1[idx] - ns1_exp)^2 / ns1_exp;
      X2_2_samp += (ltri_ns2[idx] - ns2_exp)^2 / ns2_exp;
      X2_1_rep += (ltri_ns1_tilde[idx] - ns1_exp)^2 / ns1_exp;
      X2_2_rep += (ltri_ns2_tilde[idx] - ns2_exp)^2 / ns2_exp;
    }
    X2_1_gt = X2_1_rep > X2_1_samp;
    X2_2_gt = X2_2_rep > X2_2_samp;
  }
}
```

Step 3: Prepare data and sampling

```
# Reduced frequency vector
to_ltri <- function(tab_x) {
    symm_x <- tab_x + t(tab_x) - diag(diag(tab_x))
    ltri_x <- symm_x[lower.tri(symm_x, diag = TRUE)]
}
(ltri_nmz <- to_ltri(tab_mz))
```

```
[1] 24  5  2 15  5  4
```

```
(ltri_ndz <- to_ltri(tab_dz))
```

```
[1] 14 12  6  6  6  1
```

```
ace_fit <- ace$sample(
    list(k = 3, ltri_ns1 = ltri_nmz, ltri_ns2 = ltri_ndz),
    chains = 4,
    iter_warmup = 1000,
    iter_sampling = 1000,
    seed = 732 # for reproducibility
)
```

Step 4: Check convergence

```
params <- c("rho1", "rho2", "thres", "h2", "c2", "e2")
ace_fit$print(params)
```

```
 variable  mean median   sd  mad    q5  q95 rhat ess_bulk ess_tail
 rho1      0.77   0.78 0.08 0.08  0.62 0.88 1.00     3848     2614
 rho2      0.17   0.18 0.17 0.17 -0.12 0.44 1.00     3473     2789
 thres[1]  0.03   0.03 0.10 0.10 -0.14 0.19 1.00     3414     2968
 thres[2]  1.05   1.05 0.12 0.12  0.85 1.24 1.00     4957     3011
 h2        1.19   1.19 0.37 0.37  0.60 1.82 1.00     3518     2766
 c2       -0.43  -0.42 0.34 0.34 -1.02 0.13 1.00     3488     2828
 e2        0.23   0.22 0.08 0.08  0.12 0.38 1.00     3848     2614
```

```
ace_fit$diagnostic_summary()
```

```
$num_divergent
[1] 0 0 0 0

$num_max_treedepth
[1] 0 0 0 0

$ebfmi
[1] 1.119438 0.978090 1.064659 1.107778
```

```
ace_fit$draws(params) |>
    mcmc_rank_hist()
```

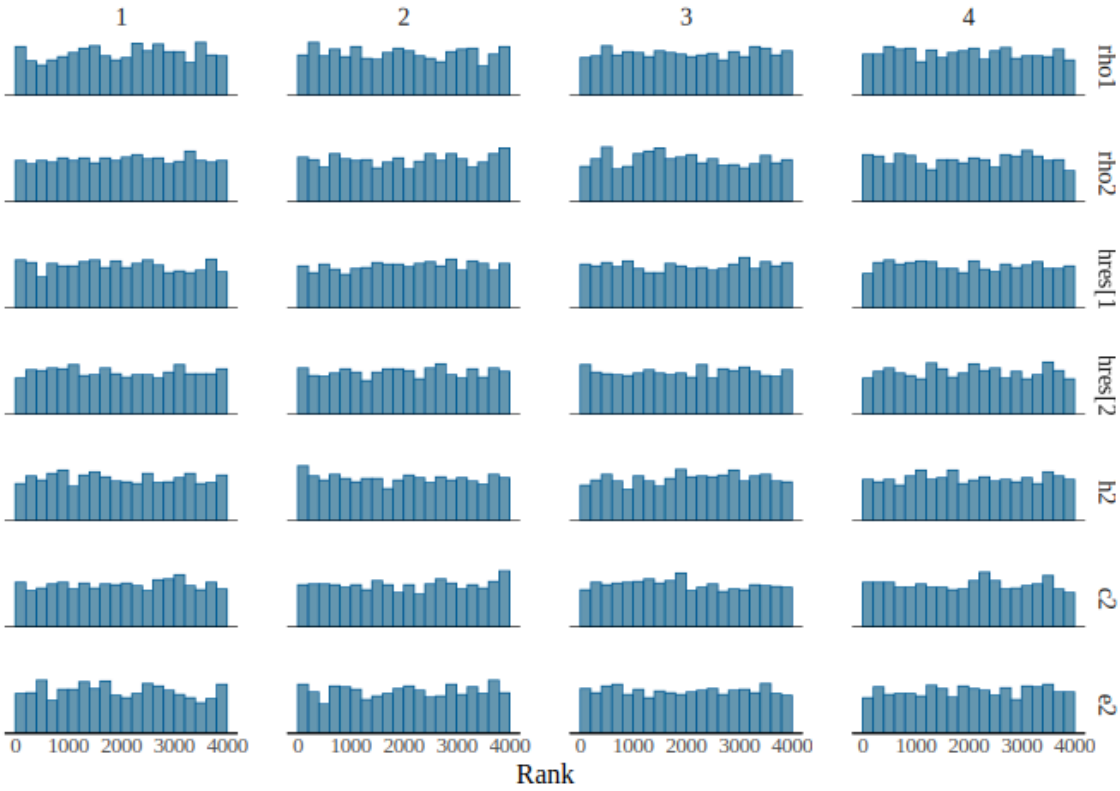


Figure 1: Figure 1: Rank histograms for the main model parameters

Step 5: Model checking

Repeat Steps 3 and 4 with reparameterized model

With unreasonable h2 and c2, we reparameterize the model to enforce theory-informed priors/ constraints.

```
//
// This Stan program estimates the polychoric correlation between two
// ordinal variables, and decompose based on an ACE model. The model is
// parameterized using a simplex for the ACE components to enforce
// nonzero variances, and assumes symmetry in the contingency table.
//
functions {
  #include "ace_fun.stan"
}
data {
  int<lower=0> k;  // number of categories
  array[k * (k + 1) %/% 2] int<lower=0> ltri_ns1;  // mz
  array[k * (k + 1) %/% 2] int<lower=0> ltri_ns2;  // dz
```

```
}
transformed data {
  int kprime = k * (k + 1) %/% 2;
  int N1 = sum(ltri_ns1);  // total N for mz
  real N1real = N1;  // convert from integer to real number
  int N2 = sum(ltri_ns2);  // total N for dz
  real N2real = N2;  // convert from integer to real number
}
parameters {
  simplex[3] ace;
  ordered[k - 1] thres;
}
transformed parameters {
  real rho1 = ace[1] + ace[2];
  real rho2 = ace[1] / 2 + ace[2];
  // Implied probabilities for each cell
  vector<lower=0>[kprime] pi1 =
    vec_symm(polycor_to_prob(thres, rho1));
  vector<lower=0>[kprime] pi2 =
    vec_symm(polycor_to_prob(thres, rho2));
}
// The model to be estimated.
model {
  // prior
  ace ~ dirichlet([1, 1, 1]');
  thres ~ student_t(4, 0, 2.5);
  // likelihood
  ltri_ns1 ~ multinomial(pi1);
  ltri_ns2 ~ multinomial(pi2);
}
generated quantities {
  // Extract the simplex parameter into a^2, c^2, and e^2
  real h2 = ace[1];
  real c2 = ace[2];
  real e2 = ace[3];
  // Intialize X^2 for each group
  real X2_1_samp = 0.0;
  real X2_2_samp = 0.0;
  // Initialize X^2 > X^2_rep indicator (for posterior predictive p)
  int<lower=0, upper=1> X2_1_gt;
  int<lower=0, upper=1> X2_2_gt;
  // Simulate new cell counts
  array[kprime] int ltri_ns1_tilde = multinomial_rng(pi1, N1);
```

```
  array[kprime] int ltri_ns2_tilde = multinomial_rng(pi2, N2);
  {
    vector[kprime] N1pi1 = N1real .* pi1;
    vector[kprime] N2pi2 = N2real .* pi2;
    // initialize X^2_rep for each group
    real X2_1_rep = 0.0;
    real X2_2_rep = 0.0;
    for (idx in 1:kprime) {
      real ns1_exp = N1pi1[idx];
      real ns2_exp = N2pi2[idx];
      // loop over each cell, and add the discrepancy to X2
      X2_1_samp += (ltri_ns1[idx] - ns1_exp)^2 / ns1_exp;
      X2_2_samp += (ltri_ns2[idx] - ns2_exp)^2 / ns2_exp;
      X2_1_rep += (ltri_ns1_tilde[idx] - ns1_exp)^2 / ns1_exp;
      X2_2_rep += (ltri_ns2_tilde[idx] - ns2_exp)^2 / ns2_exp;
    }
    X2_1_gt = X2_1_rep > X2_1_samp;
    X2_2_gt = X2_2_rep > X2_2_samp;
  }
}
```

One can then repeat steps 3 and 4 to obtain the posterior samples and convergence diagnostics, which we do not repeat the details here. The posterior summaries of the main parameters are in Table 1.

```
ace_s_fit <- ace_simplex$sample(
    list(k = 3, ltri_ns1 = ltri_nmz, ltri_ns2 = ltri_ndz),
    chains = 4,
    iter_warmup = 1000,
    iter_sampling = 1000,
    seed = 732 # for reproducibility
)
```

```
ace_s_fit$summary(params, mean, median, sd, mad, posterior::quantile2, hdi2) |>
    transform(
        variable = paste0(
            variable,
            " (",
            c(
                "$\\rho^\\text{MZ}$",
                "$\\rho^\\text{DZ}$",
                "$\\tau_1$",
```

```
                "$\\tau_2$",
                "$h^2$",
                "$c^2$",
                "$e^2$"
            ),
            ")"
        )
    ) |>
    knitr::kable(digits = 2)
```

Table 1

Table 1: Posterior summary of the main model parameters, based on the simplex parameterization

variable	mean	median	sd	mad	q5	q95	hdi_lo	hdi_hi
rho1 (ρ^{MZ})	0.75	0.76	0.09	0.09	0.58	0.87	0.61	0.89
rho2 (ρ^{DZ})	0.45	0.44	0.07	0.07	0.34	0.57	0.33	0.56
thres[1] (τ_1)	0.03	0.02	0.10	0.10	−0.14	0.20	−0.15	0.20
thres[2] (τ_2)	1.04	1.04	0.12	0.12	0.84	1.24	0.83	1.23
h2 (h^2)	0.60	0.63	0.17	0.15	0.27	0.81	0.34	0.85
c2 (c^2)	0.15	0.11	0.13	0.11	0.01	0.40	0.00	0.33
e2 (e^2)	0.25	0.24	0.09	0.09	0.13	0.42	0.11	0.39

Global fit

Local fit

```
x_scale <- scale_x_continuous(
    breaks = seq_along(ltri_nmz),
    labels = c(
        "(1,1)",
        "(2,1)+(1,2)",
        "(3,1)+(1,3)",
        "(2,2)",
        "(3,2)+(2,3)",
        "(3,3)"
    )
)
```

```
ace_s_fit$draws("ltri_ns1_tilde", format = "matrix") |>
    ppc_intervals(y = ltri_nmz) +
    xlab("Cell (twin1, twin2)") +
    x_scale
```

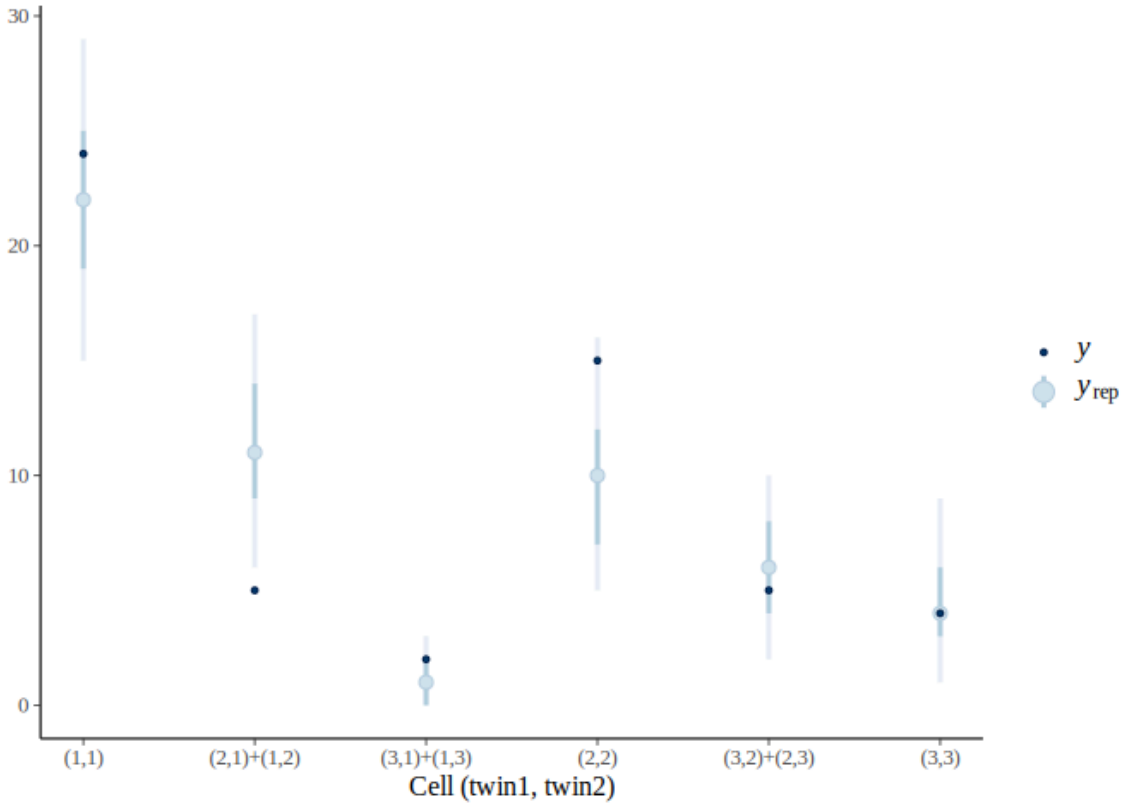


Figure 2: Figure 2: Posterior predictive check for each cell count in the contingency tables for MZ twins

```
ace_s_fit$draws("ltri_ns2_tilde", format = "matrix") |>
    ppc_intervals(y = ltri_ndz) +
    xlab("Cell (twin1, twin2)") +
    x_scale
```

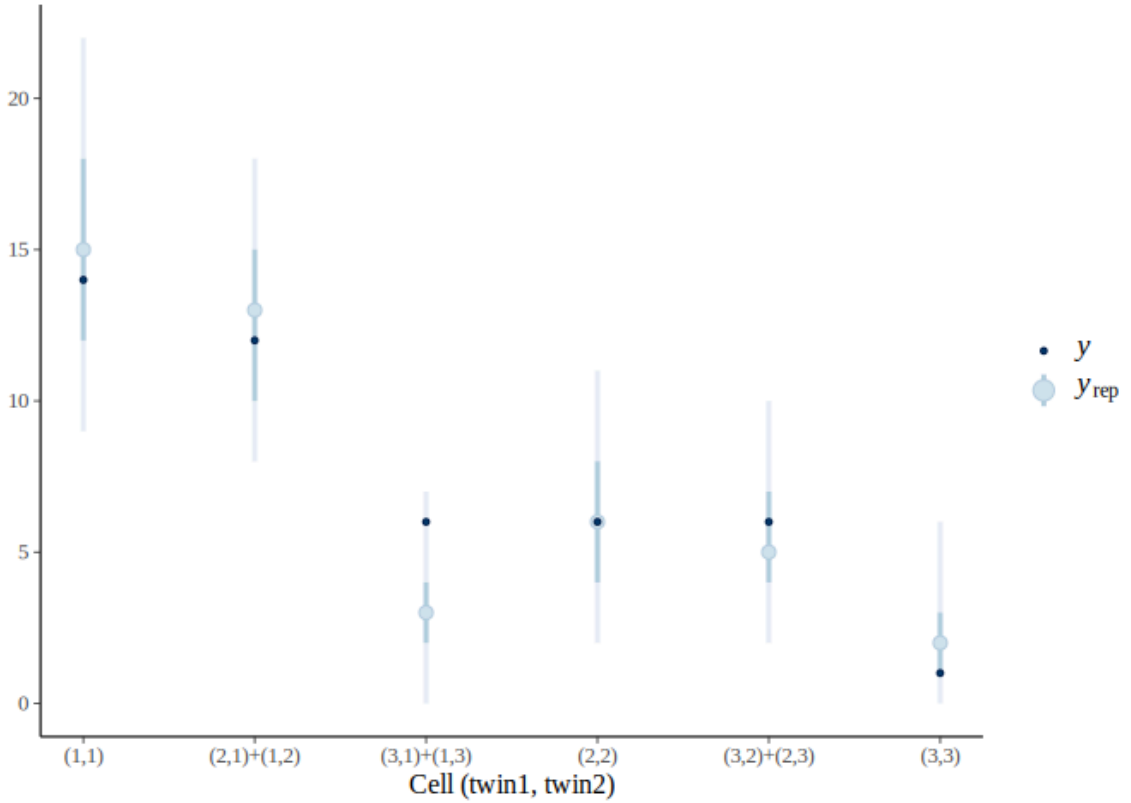


Figure 3: Figure 3: Posterior predictive check for each cell count in the contingency tables for DZ twins

```
                        # Obtain residuals
   ntilde1 <- ace_s_fit$draws("ltri_ns1_tilde", format = "draws_df")
   ntilde2 <- ace_s_fit$draws("ltri_ns2_tilde", format = "draws_df")
            # Residual: Observed counts - Predicted counts
res1 <- sweep(ntilde1[1:6], MARGIN = 2, ltri_nmz, FUN = \(x, y) y - x)
res2 <- sweep(ntilde2[1:6], MARGIN = 2, ltri_ndz, FUN = \(x, y) y - x)
                    # Posterior predictive p value
             ppp1 <- sapply(res1, function(x) mean(x < 0))
             ppp2 <- sapply(res2, function(x) mean(x < 0))

                    # Prepare data for plotting MZ
      new_tab_mz <- as.table(tab_mz) # copy to preserve original
      new_tab_mz[lower.tri(new_tab_mz, diag = TRUE)] <- ltri_nmz
               new_tab_mz[upper.tri(new_tab_mz)] <- NA
                tab_mz_df <- as.data.frame(new_tab_mz)
                    # Add ppp and residual values
             tab_mz_df$ppp[!is.na(tab_mz_df$Freq)] <- ppp1
          tab_mz_df$res[!is.na(tab_mz_df$Freq)] <- paste0(
                        round(colMeans(res1), 1),
                                " ± ",
                       round(apply(res1, 2, sd), 1),
                                "\n (",
                              round(ppp1, 2),
                                ")"
                              )

               # Reverse order of twin1 for plot purposes
                      tab_mz_df$twin1 <- factor(
                             tab_mz_df$twin1,
                  levels = rev(levels(tab_mz_df$twin1))
                              )
                          p1 <- tab_mz_df |>
               ggplot(aes(x = twin2, y = twin1, fill = ppp)) +
                        # geom_tile to draw the graph
                              geom_tile() +
                 # Add residuals directly with geom_text()
                geom_text(aes(label = res), na.rm = TRUE) +
                           # Put x axis on top
                    scale_x_discrete(position = "top") +
          # Color the graph: Red for small ppp, blue for large ppp
      # Here ppp close to zero are red, and those close to one are blue
# gradient2 instead of gradient gives us a "mid" color which we can make white
                          scale_fill_gradient2(
                                low = "red",
                               high = "blue",
                               mid = "lightgrey",
                            midpoint = 0.5,
                        na.value = rgb(0, 0, 0, 0),
                            limit = c(0, 1),
                             space = "Lab"
```

(a) MZ Twins

(b) DZ Twins

Figure 4: Figure 4: Posterior predictive check for each cell in the reduced contingency tables

Step 6: Summarizing the posterior

```
ace_s_fit$draws(params) |>
    mcmc_hist()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

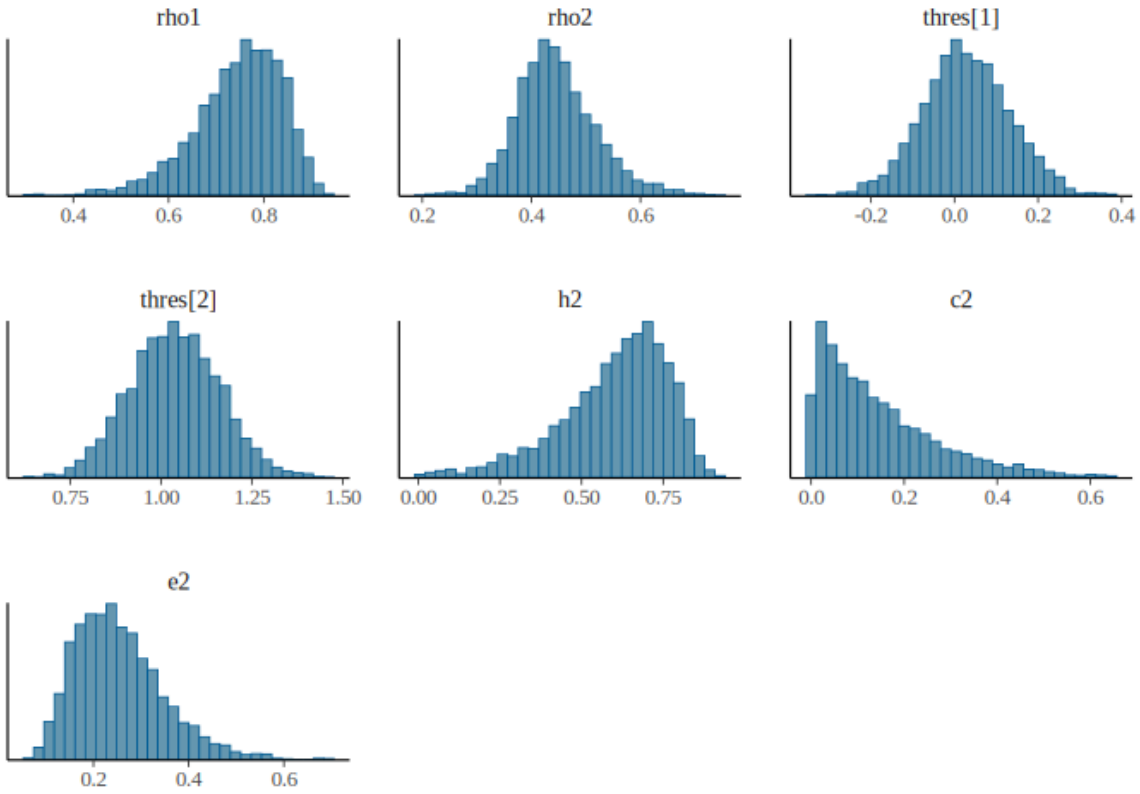


Figure 5: Figure 5: Posterior distributions of the main model parameters

Compare with *umx* package

```
library(umx)
```

```
Loading required package: OpenMx

To take full advantage of multiple cores, use:
  mxOption(key='Number of Threads', value=parallel::detectCores()) #now
  Sys.setenv(OMP_NUM_THREADS=parallel::detectCores()) #before library(OpenMx)

Registered S3 method overwritten by 'quantmod':
  method            from
```

```
  as.zoo.data.frame zoo

For an overview type '?umx'

Attaching package: 'umx'

The following object is masked from 'package:stats':

    loadings
```

```
# Make the ordinal variables into umxFactors
dat[, c("twin1", "twin2")] <- umxFactor(dat[, c("twin1", "twin2")])

mz_data <- dat[dat$zygosity == "MZ", ]
dz_data <- dat[dat$zygosity == "DZ", ]

# Model and summary with FIML
m1 <- umxACE(
    selDVs = "twin",
    dzData = dz_data,
    mzData = mz_data,
    sep = "",
    tryHard = "ordinal"
)
```

```
Found 1 pair(s) of ordinal variables:'twin1' and 'twin2' (No binary)

Running ACE with 4 parameters

Beginning initial fit attempt

Running ACE with 4 parameters

 Lowest minimum so far:  365.978829144973

Beginning fit attempt 1 of at maximum 10 extra tries

Running ACE with 4 parameters

Beginning fit attempt 2 of at maximum 10 extra tries

Running ACE with 4 parameters
```

```
 Lowest minimum so far:  365.978829144971
Beginning fit attempt 3 of at maximum 10 extra tries
Running ACE with 4 parameters
 Lowest minimum so far:  365.978829144971
Beginning fit attempt 4 of at maximum 10 extra tries
Running ACE with 4 parameters
 Lowest minimum so far:  365.97882914497
Beginning fit attempt 5 of at maximum 10 extra tries
Running ACE with 4 parameters
Beginning fit attempt 6 of at maximum 10 extra tries
Running ACE with 4 parameters
Beginning fit attempt 7 of at maximum 10 extra tries
Running ACE with 4 parameters
Beginning fit attempt 8 of at maximum 10 extra tries
Running ACE with 4 parameters
Beginning fit attempt 9 of at maximum 10 extra tries
Running ACE with 4 parameters
Beginning fit attempt 10 of at maximum 10 extra tries
Running ACE with 4 parameters
Retry limit reached
Solution found
```

```
Final run, for Hessian and/or standard errors and/or confidence intervals

Running ACE with 4 parameters

 Solution found!  Final fit=365.97883 (started at 370.92682)  (11 attempt(s):
11 valid, 0 errors)

 Start values from best fit:

-0.024758927371081,0.934118071479837,0,0.471878643174057

ACE -2 × log(Likelihood) = 365.979

Table: Standardized parameter estimates from a 1-factor Cholesky ACE model. A:
additive genetic; C: common environment; E: unique environment.

|     |    a1|c1 |    e1|
|:----|-----:|:--|-----:|
|twin | 0.893|.  | 0.451|


Table: Means (from model$top$expMean)

|          |  twin1|  twin2|
|:---------|------:|------:|
|intercept | -0.025| -0.025|

?umxPlotACE options: std=T/F, means=T/F, digits=n, strip_zero=T/F, file=, min=,
max =
```

```
umxSummary(m1)
```

```
ACE -2 × log(Likelihood) = 365.98

Table: Standardized parameter estimates from a 1-factor Cholesky ACE model. A:
additive genetic; C: common environment; E: unique environment.

|     |   a1|c1 |   e1|
|:----|----:|:--|----:|
|twin | 0.89|.  | 0.45|
```

```
Table: Means (from model$top$expMean)

|          |  twin1|  twin2|
|:---------|------:|------:|
|intercept | -0.025| -0.025|

?umxPlotACE options: std=T/F, means=T/F, digits=n, strip_zero=T/F, file=, min=,
max =
```

```
# CI failed
umxConfint(m1, run = TRUE)
```

```
Running ACE with 4 parameters

               lbound estimate ubound lbound Code ubound Code
top.a_std[1,1]     NA    0.893     NA          NA          NA
top.c_std[1,1]     NA    0.000  0.577          NA           3
top.e_std[1,1]     NA    0.451     NA          NA          NA
```

```
# Model and summary with DWLS
m1_dwls <- umxACE(
    selDVs = "twin",
    dzData = dz_data,
    mzData = mz_data,
    sep = "",
    type = "DWLS",
    tryHard = "ordinal"
)
```

```
Found 1 pair(s) of ordinal variables:'twin1' and 'twin2' (No binary)

Data treated as DWLS

Running ACE with 4 parameters

Beginning initial fit attempt

Running ACE with 4 parameters

 Lowest minimum so far:  0.0391834784698901
```

```
Beginning fit attempt 1 of at maximum 10 extra tries

Running ACE with 4 parameters

 Lowest minimum so far:  0.0391834784698893

Beginning fit attempt 2 of at maximum 10 extra tries

Running ACE with 4 parameters

 Lowest minimum so far:  0.0391834784698892

Beginning fit attempt 3 of at maximum 10 extra tries

Running ACE with 4 parameters

Beginning fit attempt 4 of at maximum 10 extra tries

Running ACE with 4 parameters

Beginning fit attempt 5 of at maximum 10 extra tries

Running ACE with 4 parameters

 Lowest minimum so far:  0.0391834784698892

Beginning fit attempt 6 of at maximum 10 extra tries

Running ACE with 4 parameters

Beginning fit attempt 7 of at maximum 10 extra tries

Running ACE with 4 parameters

Beginning fit attempt 8 of at maximum 10 extra tries

Running ACE with 4 parameters

 Lowest minimum so far:  0.0391834784698892

Beginning fit attempt 9 of at maximum 10 extra tries

Running ACE with 4 parameters
```

```
Beginning fit attempt 10 of at maximum 10 extra tries

Running ACE with 4 parameters

Retry limit reached

Solution found

Final run, for Hessian and/or standard errors and/or confidence intervals

Running ACE with 4 parameters

 Solution found!  Final fit=0.039183478 (started at 0.10351156)  (11 attempt(s):
11 valid, 0 errors)

 Start values from best fit:

-8.39772023278525e-05,0.895116918606286,9.4084489021653e-09,0.461643857432493

ACE -2 × log(Likelihood) = 6.478

Table: Standardized parameter estimates from a 1-factor Cholesky ACE model. A:
additive genetic; C: common environment; E: unique environment.

|     |    a1|c1 |    e1|
|:----|-----:|:--|-----:|
|twin | 0.889|.  | 0.458|


Table: Means (from model$top$expMean)

|          |twin1 |twin2 |
|:---------|:-----|:-----|
|intercept |0     |0     |

?umxPlotACE options: std=T/F, means=T/F, digits=n, strip_zero=T/F, file=, min=,
max =
```

```
umxSummary(m1_dwls)
```

```
ACE -2 × log(Likelihood) = 6.48

Table: Standardized parameter estimates from a 1-factor Cholesky ACE model. A:
additive genetic; C: common environment; E: unique environment.

|     |   a1|c1 |   e1|
|:----|----:|:--|----:|
|twin | 0.89|.  | 0.46|


Table: Means (from model$top$expMean)

|          |twin1 |twin2 |
|:---------|:-----|:-----|
|intercept |0     |0     |

?umxPlotACE options: std=T/F, means=T/F, digits=n, strip_zero=T/F, file=, min=,
max =
```

```
# Extremely large standard errors
summary(m1_dwls)
```

```
Summary of ACE

free parameters:
           name      matrix   row   col      Estimate    Std.Error A lbound
1 expMean_twin1 top.expMean means twin1 -8.397720e-05 1.974439e+00
2        a_r1c1       top.a     1     1  8.951169e-01 1.039425e+00        0
3        c_r1c1       top.c     1     1  9.408449e-09 1.122410e-08 !     0!
4        e_r1c1       top.e     1     1  4.616439e-01 8.338411e-01        0
  ubound
1
2
3
4

To obtain confidence intervals re-run with intervals=TRUE

Model Statistics:
               |  Parameters  |  Degrees of Freedom  |  Fit (r'wr units)
       Model:              4                     6                  NA
   Saturated:             NA                    NA                  NA
```

```
Independence:            NA                    NA                    NA
Number of observations/statistics: 100/10

chi-square:  χ² ( df=NA ) = 6.477706,  p = 0.3718542
CFI: NA
TLI: NA   (also known as NNFI)
RMSEA:  0.02821661  [95% CI (0, 0.1500864)]
Prob(RMSEA <= 0.05): 0.5253774
To get additional fit indices, see help(mxRefModels)
timestamp: 2026-10-14 08:34:52
Wall clock time: 0.1118538 secs
optimizer:  SLSQP
OpenMx version number: 2.22.11
Need help?  See help(mxSummary)
```

```r
# Model and summary with WLS
m1_wls <- umxACE(
    selDVs = "twin",
    dzData = dz_data,
    mzData = mz_data,
    sep = "",
    type = "WLS",
    tryHard = "ordinal"
)
```

```
Found 1 pair(s) of ordinal variables:'twin1' and 'twin2' (No binary)

Data treated as WLS

Running ACE with 4 parameters

Beginning initial fit attempt

Running ACE with 4 parameters

 Lowest minimum so far:  0.078817106554107

Beginning fit attempt 1 of at maximum 10 extra tries

Running ACE with 4 parameters

Beginning fit attempt 2 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters

Beginning fit attempt 3 of at maximum 10 extra tries

Running ACE with 4 parameters

 Lowest minimum so far:  0.0788171065541069

Beginning fit attempt 4 of at maximum 10 extra tries

Running ACE with 4 parameters

Beginning fit attempt 5 of at maximum 10 extra tries

Running ACE with 4 parameters

 Lowest minimum so far:  0.0788171065541069

Beginning fit attempt 6 of at maximum 10 extra tries

Running ACE with 4 parameters

Beginning fit attempt 7 of at maximum 10 extra tries

Running ACE with 4 parameters

Beginning fit attempt 8 of at maximum 10 extra tries

Running ACE with 4 parameters

Beginning fit attempt 9 of at maximum 10 extra tries

Running ACE with 4 parameters

Beginning fit attempt 10 of at maximum 10 extra tries

Running ACE with 4 parameters

Retry limit reached

Solution found
```

```
Final run, for Hessian and/or standard errors and/or confidence intervals

Running ACE with 4 parameters

 Solution found!  Final fit=0.078817107 (started at 0.20654351)  (11 attempt(s):
11 valid, 0 errors)

 Start values from best fit:

-0.0342164605804386,0.971005705861823,0,0.447841878956487

ACE -2 × log(Likelihood) = 7.882

Table: Standardized parameter estimates from a 1-factor Cholesky ACE model. A:
additive genetic; C: common environment; E: unique environment.

|     |    a1|c1 |    e1|
|:----|-----:|:--|-----:|
|twin | 0.908|.  | 0.419|


Table: Means (from model$top$expMean)

|          |  twin1|  twin2|
|:---------|------:|------:|
|intercept | -0.034| -0.034|

?umxPlotACE options: std=T/F, means=T/F, digits=n, strip_zero=T/F, file=, min=,
max =
```

```
umxSummary(m1_wls)
```

```
ACE -2 × log(Likelihood) = 7.88

Table: Standardized parameter estimates from a 1-factor Cholesky ACE model. A:
additive genetic; C: common environment; E: unique environment.

|     |   a1|c1 |   e1|
|:----|----:|:--|----:|
|twin | 0.91|.  | 0.42|
```

```
Table: Means (from model$top$expMean)

|          |  twin1|  twin2|
|:---------|------:|------:|
|intercept | -0.034| -0.034|

?umxPlotACE options: std=T/F, means=T/F, digits=n, strip_zero=T/F, file=, min=,
max =
```

```
# CI failed
umxConfint(m1_wls, run = TRUE)
```

```
Running ACE with 4 parameters

               lbound estimate ubound lbound Code ubound Code
top.a_std[1,1]     NA    0.908     NA          NA          NA
top.c_std[1,1]     NA    0.000     NA          NA          NA
top.e_std[1,1]     NA    0.419     NA          NA          NA
```